Exploratory Data Analysis on the UsArrests Dataset

Report

Introduction

The US Arrests dataset contains statistics for all 50 U.S. states on arrests per 100,000 residents for murder, assault, and rape, along with the percentage of the population living in urban areas. The goal of this exploratory data analysis (EDA) is to understand relationships between these variables, identify trends and outliers, and draw conclusions about crime patterns across the United States.

Dataset Description

The dataset consists of 50 observations, each representing a U.S. state. Four numerical variables are included:

- **Murder:** Number of murders per 100,000 people.
- **Assault:** Number of assaults per 100,000 people.
- **UrbanPop:** Percentage of the population living in urban areas.
- **Rape:** Number of rapes per 100,000 people.

Each observation represents aggregated crime statistics for one state.

Data Cleaning

Before analysing the dataset, several preprocessing steps were performed.

- The dataset was successfully loaded into a pandas DataFrame.
- Column names and data types were inspected.
- Duplicate observations were checked.
- Summary statistics were generated to better understand the variables.

No duplicate records were identified, and all numerical columns were stored using the correct data types.

Missing Data

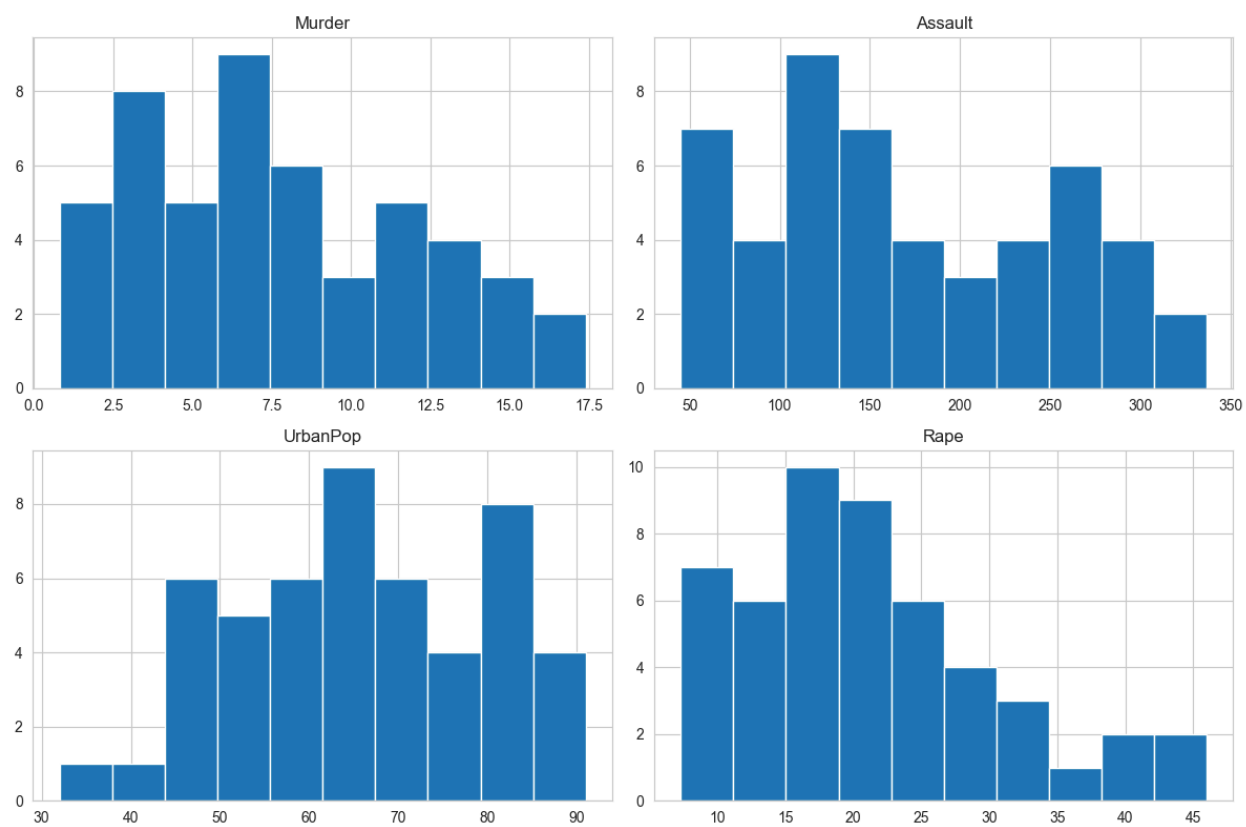
The dataset was examined for missing values before beginning the analysis.

No missing values were identified in any of the variables. Consequently, no rows needed to be removed, and no imputation techniques were required.

Data Stories and Visualisations

Exploratory Data Analysis

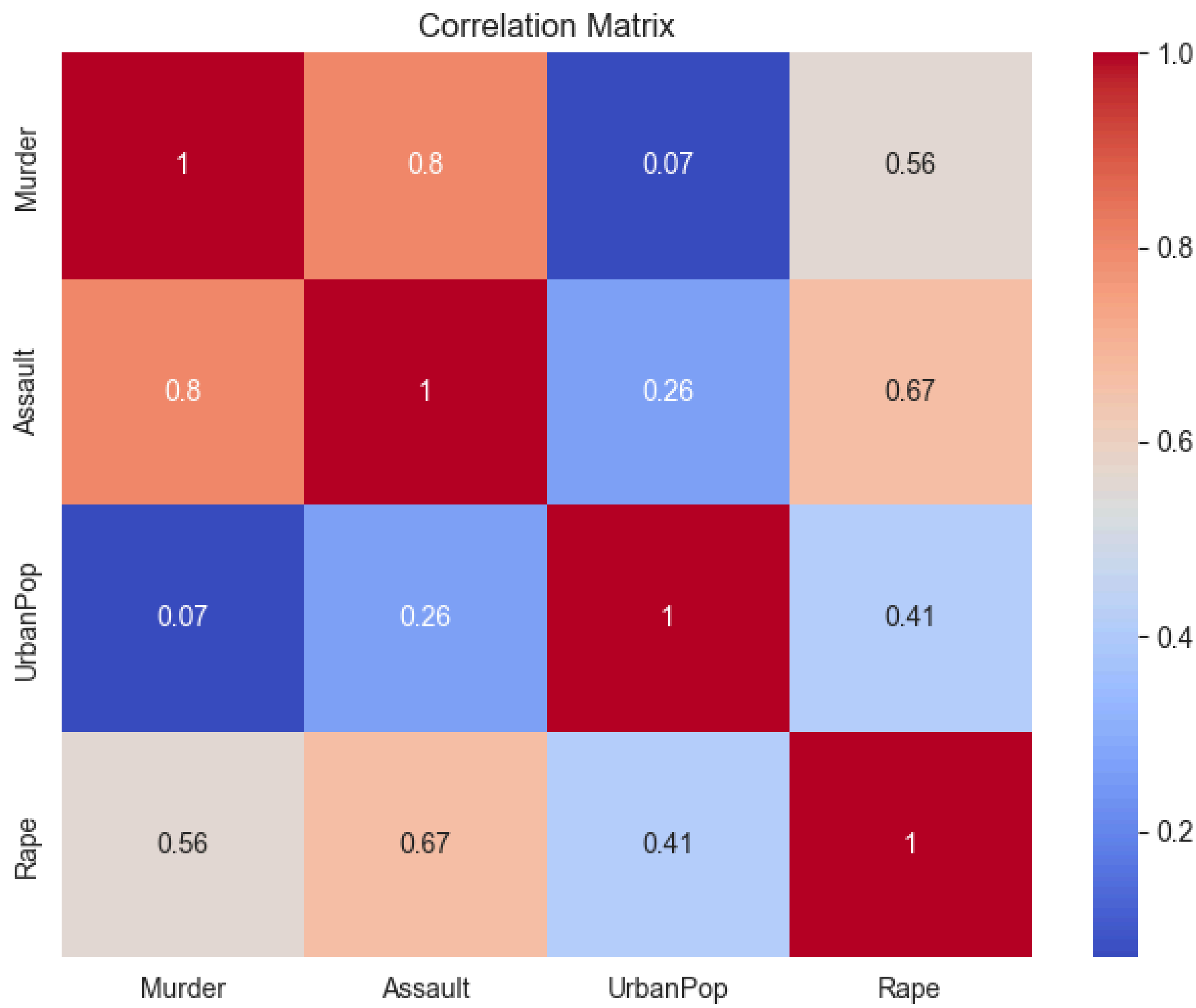
Figure 1: Distribution of Crime Rates



Histograms were created to visualise the distribution of each variable.

The histograms indicate that several crime variables are right-skewed, suggesting that while most states have relatively moderate crime rates, a small number experience substantially higher crime rates. Urban population appears to be more evenly distributed across states.

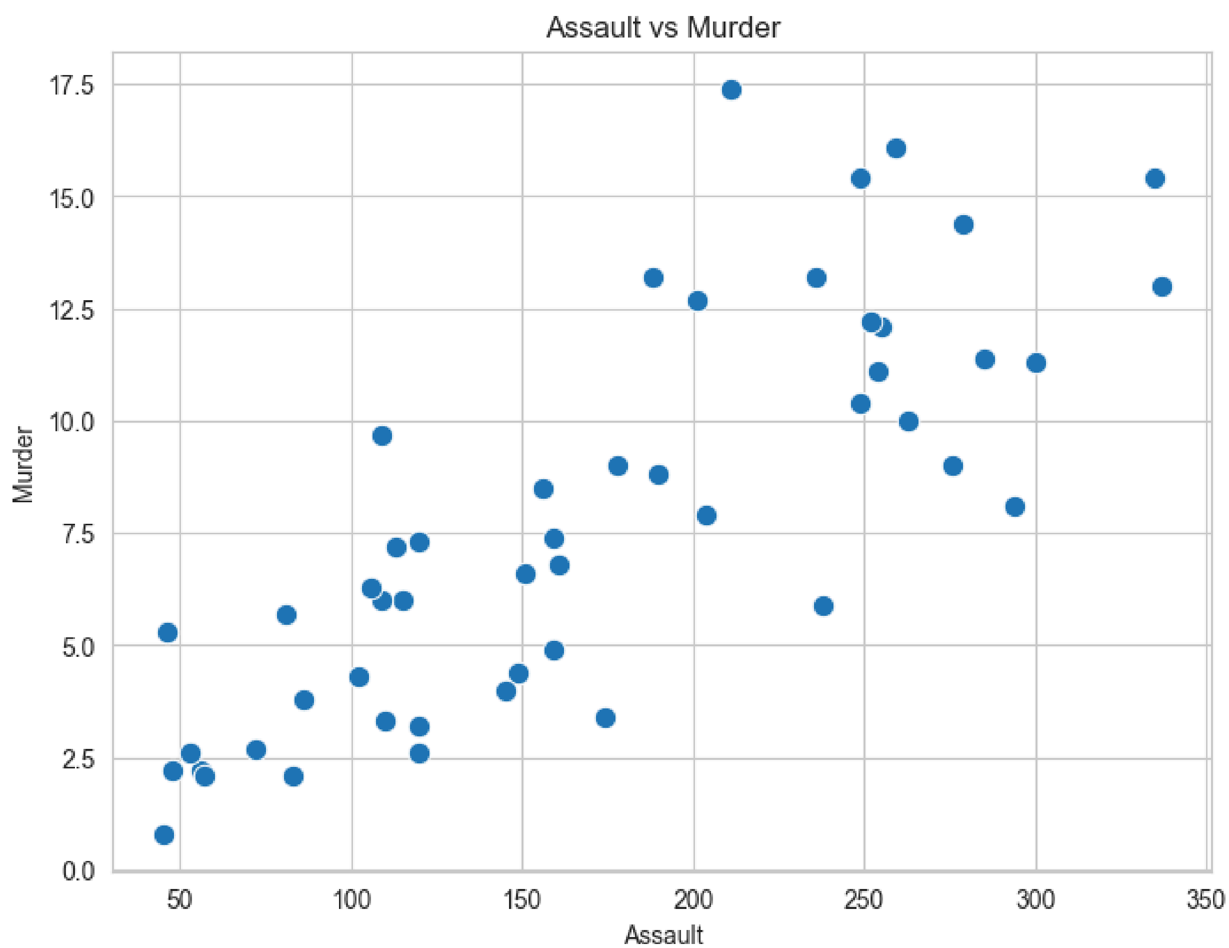
Figure 2: Correlation Matrix



A correlation heatmap was produced to examine the relationships between the numerical variables.

The heatmap reveals a strong positive correlation between murder and assault rates. This indicates that states with high assault rates generally also experience higher murder rates. Urban population exhibits only weak correlations with the crime variables.

Figure 3: Assault vs. Murder



A scatterplot was created to investigate the relationship between assault and murder rates.

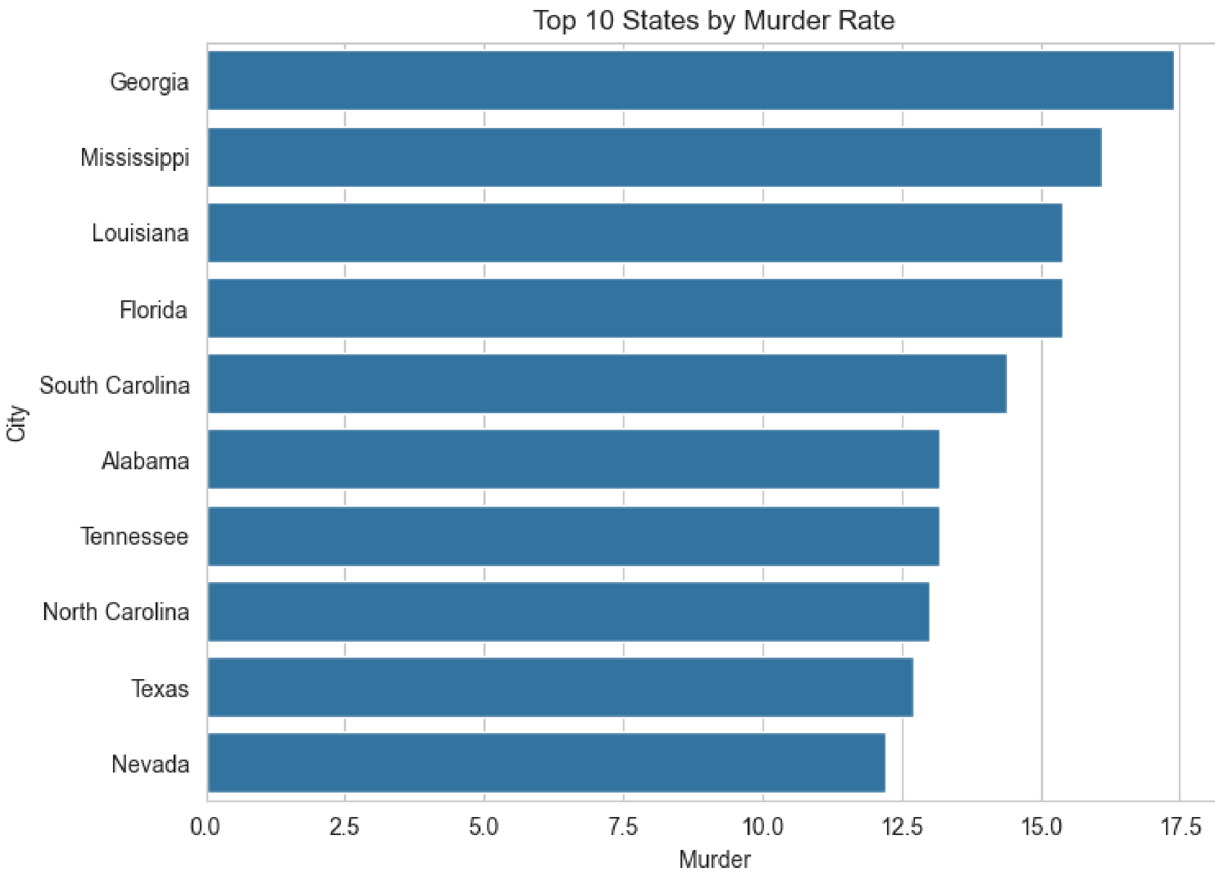
The scatterplot demonstrates a clear positive relationship between assault and murder rates. States with higher assault rates tend to have higher murder rates, supporting the findings observed in the correlation matrix.

Figure 4: Urban Population vs. Murder

A regression plot was used to examine whether urban population is associated with murder rates.

The regression line suggests only a weak relationship between urban population and murder rates. Although some highly urbanised states exhibit elevated murder rates, urbanisation alone does not appear to be a strong predictor of violent crime.

Figure 4: Top 10 States by Murder Rate



A horizontal bar chart was created to display the ten states with the highest murder rates.

This chart clearly identifies the states with the highest murder rates and allows for easy comparison. The differences between the highest-ranking states and the remainder of the dataset are clearly visible, highlighting the variation in violent crime across the United States.

Key Findings

The exploratory data analysis identified several important findings:

- Assault and murder exhibit a strong positive relationship.
- Urban population has only a weak relationship with violent crime.
- Several states appear as statistical outliers due to exceptionally high crime rates.
- The dataset contained no missing values or duplicate observations.
- The visualisations consistently support the conclusion that violent crime measures tend to increase together.

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